



Course Details

Course Title: Quantum Mechanics

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Submitted To

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Topic: Blackbody

1: Sunspots appear dark, although their temperature are typically 5000K because the rest of the Sun's surface is even hotter, about 5800K. Compare the radiation rates of surfaces of the same emissivity whose temperature are respectively 5000 and 5800K.

Soln:

Given that,

Different temperatures are $T_1 = 5000\text{K}$ and $T_2 = 5800\text{K}$

We know that,

$$E = aT^4 \quad [\text{Stefan - Boltzmann Law}]$$

Where a is the constant and T is temperature.

$\therefore$ The radiation rates of the different temperature will be,

$$\begin{aligned} \frac{E_1}{E_2} &= \frac{aT_1^4}{aT_2^4} \\ &= \frac{T_1^4}{T_2^4} \\ &= \frac{(5000)^4}{(5800)^4} \\ &= 0.552 \end{aligned}$$

$$\therefore \frac{E_1}{E_2} = 0.552$$

$$\text{or, } E_1 = 0.552 E_2$$

$\therefore$ The radiation rate of the sunspot at 5000K is approximately 0.552 times of the surrounding surface at 5800K.

2. At what rate would solar energy arrive at the earth if the solar surface had a temperature 10 percent lower than it is?

Soln:

Given data,

$$T_2 = T_1 - 0.10T_1 \\ = 0.90T_1$$

we know,

$$E = \sigma T^4$$

Now,

$$\frac{E_2}{E_1} = \frac{\sigma T_2^4}{\sigma T_1^4} \\ \text{or, } \frac{E_2}{E_1} = \left(\frac{T_2}{T_1}\right)^4 \\ = \left(\frac{0.90T_1}{T_1}\right)^4 \\ = 0.656$$

$$\therefore E_2 = 0.656 E_1$$

This means the new rate of solar energy arriving at earth would be approximately 65.6% of the current rate.

3. An Object is at a temperature of 400°C . At what temperature would it radiate energy twice as fast?

Soln:

Given that,

$$\begin{aligned}\text{Temperature of the object, } T_1 &= 400^{\circ}\text{C} \\ &= (400 + 273.15)\text{K} \\ &= 673.15\text{K}\end{aligned}$$

This object radiates energy twice so,

$$E_2 = 2E_1$$

We know,

$$\text{The rate of radiance energy, } E = aT^4$$

$$\text{Now, } \frac{E_2}{E_1} = \frac{aT_2^4}{aT_1^4}$$

$$\text{or, } \frac{2E_1}{E_1} = \frac{T_2^4}{T_1^4}$$

$$\text{or, } 2^{\frac{1}{4}} = \frac{T_2}{T_1}$$

$$\text{or, } T_2 = (2^{\frac{1}{4}}) \times T_1$$

$$= (2^{\frac{1}{4}}) \times 673.15\text{K}$$

$$= 800.51\text{K}$$

$$= (800.51 - 273.15)^{\circ}\text{C}$$

$$= 527.36^{\circ}\text{C}$$

At 527.36°C temperature that object would radiate energy twice as fast.

4. Find the surface area of a blackbody that radiates 1.00 kW when the temperature is 500°C . If the blackbody is a sphere, what is its radius?

Soln:

Given that,

The surface area of a blackbody that radiates

$$P = 1.00 \text{ kW} \\ = 1000 \text{ W}$$

Here, Temperature, $T = 500^{\circ}\text{C} = (500 + 273.15) \text{ K}$
 $= 773.15 \text{ K}$

$$\therefore \text{Rate of radiation, } E = \sigma T^4 \\ = 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4} \times (773.15 \text{ K})^4 \\ = 2.02 \times 10^4 \text{ W m}^{-2}$$

Now,

$$\text{The rate of radiation, } E = \frac{\text{Power output}}{\text{Surface area}}$$

$$\text{or, } E = \frac{P}{4\pi r^2}$$

$$\text{or, } r = \sqrt{\frac{P}{4\pi E}}$$

$$= \sqrt{\frac{1000 \text{ W}}{4 \times 3.1416 \times 2.02 \times 10^4}}$$

$$= 0.0627 \text{ m}$$

$$\text{or, } r = 6.27 \text{ cm}$$

$\therefore$ The radius is, 6.27 cm

5. Find the peak wavelength in the spectrum of the radiation from a blackbody at a temperature of 500°C . In what part of the electromagnetic spectrum is this wavelength?

Soln:

Given that,

$$\begin{aligned}\text{Temperature, } T &= 500^{\circ}\text{C} \\ &= (500 + 273.15) \text{ K} \\ &= 773.15 \text{ K}\end{aligned}$$

We know that,

$$\lambda_{\text{max}} T = 2.898 \times 10^{-3} \text{ mk}$$

Now,

$$\begin{aligned}\lambda_{\text{max}} &= \frac{2.898 \times 10^{-3} \text{ mk}}{T} \\ &= \frac{2.898 \times 10^{-3} \text{ mk}}{773.15 \text{ K}} \\ &= 3.75 \times 10^{-6} \text{ m}\end{aligned}$$

$$= 3750 \text{ nm} \quad [\because 1 \text{ m} = 10^9 \text{ nm}]$$

$\therefore$ A wavelength of 3750 nm in the spectrum, radiates from a blackbody at a temperature of 500°C or 773.15 K .

6. The brightest part of the spectrum of the star Sirius is located at a wavelength of about 290 nm. What is the surface temperature of Sirius?

Soln:

Given that,

Wavelength, $\lambda_{\max} = 290 \text{ nm}$

$$= 2.9 \times 10^{-7} \text{ m} \quad [\text{As } 1 \text{ nm} = 10^{-9} \text{ m}]$$

Now,

$$\lambda_{\max} T = 2.898 \times 10^{-3} \text{ mK}$$

$$\text{or, } T = \frac{2.898 \times 10^{-3} \text{ mK}}{\lambda_{\max}}$$

$$= \frac{2.898 \times 10^{-3} \text{ mK}}{2.9 \times 10^{-7} \text{ m}}$$

$$= 9993.10 \text{ K}$$

$\therefore$ Surface temperature of Sirius is,
9993.10 K.

Topic: Photo - electric Effect

1. The maximum wavelength for photoelectric emission in tungsten is 230nm . What wavelength of light must be used in order for electrons with maximum energy of 1.5eV to be ejected?

Soln:

Given that,

$$K.E_{\text{max}} = 1.5\text{eV}$$

$$\begin{aligned}\text{Wavelength of tungsten, } \lambda_0 &= 230\text{nm} \\ &= 230 \times 10^{-9}\text{m}\end{aligned}$$

Now,

$$K.E_{\text{max}} = h\nu - h\nu_0$$

$$\text{or, } K.E_{\text{max}} = h(\nu - \nu_0)$$

$$\text{or, } \frac{K.E_{\text{max}}}{h} = c\left(\frac{1}{\lambda} - \frac{1}{\lambda_0}\right) \quad \left[\text{As } \nu = \frac{c}{\lambda}\right]$$

$$\text{or, } \frac{1.5 \times 1.6 \times 10^{-19}\text{J}}{6.626 \times 10^{-34}\text{Js}} = \frac{3 \times 10^8\text{ms}^{-1}}{\lambda} - \frac{3 \times 10^8\text{ms}^{-1}}{230 \times 10^{-9}\text{m}}$$

$$\text{or, } \lambda = 1.8 \times 10^{-7}\text{m}$$

$$\text{or, } \lambda = 180\text{nm}$$

$\therefore$ The wavelength of the light is, $\lambda = 180\text{nm}$

2. The minimum frequency for photoelectric emission in copper is 1.1×10^{15} Hz. Find the maximum energy of the photoelectrons (in electronvolts) when light of frequency 1.5×10^{15} Hz is directed on a copper surface?

Soln:

Given that,

minimum frequency, $\nu_0 = 1.1 \times 10^{15}$ Hz

Light of frequency, $\nu = 1.5 \times 10^{15}$ Hz

We know that,

$$K.E = h(\nu - \nu_0)$$

$$= 6.626 \times 10^{-34} \text{ Js} \times (1.5 \times 10^{15} - 1.1 \times 10^{15})$$

$$= 2.65 \times 10^{-19} \text{ J}$$

$$= \frac{2.65 \times 10^{-19}}{1.6 \times 10^{-19}} \text{ eV}$$

$$= 1.66 \text{ eV}$$

$\therefore$ The maximum energy of photoelectron,

$$K.E = 1.66 \text{ eV}$$

3 What is the maximum wavelength of light that will cause photoelectrons to be emitted from sodium? What will the maximum kinetic energy of the photoelectrons be if 200nm light fall on a sodium surface? (work function for sodium is 2.3eV).

Solⁿ:

Given that,

Work function for sodium, $\phi = 2.3\text{eV}$

We know,

$$\phi = \frac{hc}{\lambda_{\max}}$$

$$\text{or, } \lambda_{\max} = \frac{hc}{\phi} = \frac{6.626 \times 10^{-34} \text{ Js} \times 3 \times 10^8 \text{ ms}^{-1}}{2.3 \times 1.6 \times 10^{-19} \text{ J}} \\ = 5.40 \times 10^{-7} \text{ m} \\ = 540 \text{ nm}$$

$\therefore$ The maximum wavelength of light, $\lambda_{\max} = 540 \text{ nm}$

Again, $K.E_{\max} = h\nu - \phi$

$$= h \frac{c}{\lambda} - \phi$$

$$= \frac{6.626 \times 10^{-34} \text{ Js} \times 3 \times 10^8 \text{ ms}^{-1}}{200 \times 10^{-9} \text{ nm}} - 2.3 \times 1.6 \times 10^{-19} \text{ J}$$

$$= 6.26 \times 10^{-19} \text{ J}$$

$$= \frac{6.26 \times 10^{-19} \text{ J}}{1.6 \times 10^{-19} \text{ J}}$$

$$= 3.91 \text{ eV}$$

$\therefore$ The maximum kinetic energy of photoelectron 3.91eV.

4. A Metal surface illuminated by $8.5 \times 10^{14} \text{ Hz}$ light emits electrons whose maximum energy is 0.52 eV . The same surface illuminated by $12.0 \times 10^{14} \text{ Hz}$ light emits electrons whose maximum energy is 1.97 eV . From these data find Planck's constant and the work function of the surface.

Soln:

Given that,

$$\nu_1 = 8.5 \times 10^{14} \text{ Hz}$$

$$\nu_2 = 12 \times 10^{14} \text{ Hz}$$

$$K.E_{\text{max}_1} = 0.52 \text{ eV and } K.E_{\text{max}_2} = 1.97 \text{ eV}$$

Now,

$$K.E_{\text{max}_1} = h\nu_1 - h\nu_0$$

$$\text{or, } h\nu_1 = K.E_{\text{max}_1} + h\nu_0 \quad \dots \dots (i)$$

And,

$$K.E_{\text{max}_2} = h\nu_2 - h\nu_0$$

$$\text{or, } h\nu_2 = K.E_{\text{max}_2} + h\nu_0 \quad \dots \dots (ii)$$

$$(ii) - (i) \Rightarrow$$

$$h\nu_2 - h\nu_1 = K.E_{\text{max}_2} - K.E_{\text{max}_1}$$

$$\text{or, } h(12 \times 10^{14} - 8.5 \times 10^{14}) = \{ (1.97 \times 1.6 \times 10^{-19}) - (0.52 \times 1.6 \times 10^{-19}) \}$$

$$\text{or, } h = 6.628 \times 10^{-34} \text{ Js}$$

Again,

$$K.E_{\text{max}_1} = h\nu_1 - h\nu_0$$

$$\text{or, } h\nu_1 = K.E_{\text{max}_1} + h\nu_0$$

$$\text{or, } h\nu_0 = h\nu_1 - k \cdot E_{\text{max}_1}$$

$$= (6.628 \times 10^{-34} \text{ Js} \times 8.5 \times 10^{14}) - 0.52 \times 1.6 \times 10^{-19}$$

$$= 4.8 \times 10^{-19} \text{ J}$$

$$= \frac{4.8 \times 10^{-19}}{1.6 \times 10^{-19}} \text{ eV}$$

$$= 3 \text{ eV}$$

$\therefore$ Work function, $h\nu_0 = 3 \text{ eV}$.

5. The work function of a tungsten surface is 5.4 eV . When the surface is illuminated by light of wavelength 175 nm , the maximum photoelectron energy is 1.7 eV . Find the planck's constant from these data.

Soln: Given here,

Work function, $h\nu_0 = 5.4 \text{ eV}$

Wavelength, $\lambda = 175 \text{ nm}$

and maximum energy, $k \cdot E_{\text{max}} = 1.7 \text{ eV}$

We know that,

$$k \cdot E_{\text{max}} = h\nu - h\nu_0$$

$$\text{or, } (1.7 \times 1.6 \times 10^{-19} \text{ J}) = h \times \frac{3 \times 10^8 \text{ ms}^{-1}}{175 \times 10^{-9} \text{ m}} - (5.4 \times 1.6 \times 10^{-19} \text{ J})$$

$$\text{or, } h = \frac{1.136 \times 10^{-18}}{1.714 \times 10^{15}}$$

$$= 6.628 \times 10^{-34} \text{ Js}$$

Topic: Compton Effect

1. A monochromatic X-ray beam whose wavelength is 55.8 pm is scattered through 46° . Find the wavelength of the scattered beam.

Soln: Given here,

$$\text{Initial wavelength, } \lambda = 55.8 \text{ pm} \\ = 55.8 \times 10^{-12} \text{ m}$$

$$\text{Scattering angle, } \phi = 46^\circ$$

$$\text{Mass of electron, } m = 9.1 \times 10^{-31} \text{ kg}$$

We know,

Compton shift of wavelength,

$$\lambda' - \lambda = \lambda_c (1 - \cos \phi)$$

$$\text{or, } \lambda' - \lambda = \frac{h}{mc} (1 - \cos \phi)$$

$$\text{or, } \lambda' = \lambda + \frac{h}{mc} (1 - \cos \phi)$$

$$= 55.8 \times 10^{-12} \text{ m} + \frac{6.626 \times 10^{-34} \text{ Js}}{9.1 \times 10^{-31} \text{ kg} \times 3 \times 10^8 \text{ ms}^{-1}} (1 - \cos 46^\circ) \\ = 5.65 \times 10^{-11} \text{ m} \\ = 56.5 \text{ pm}$$

$\therefore$ Wavelength of the scattered beam is,
 56.5 pm

2. A beam of X-ray is scattered by a target. At 45° from the beam direction the scattered X-rays have a wavelength of $2.2 \text{ \AA}$. What is the wavelength of the X-rays in the direction beam?

Soln:

Given that,

Scattered X-rays wavelength, $\lambda' = 2.2 \text{ \AA}$
 $= 2.2 \times 10^{-12} \text{ m}$

Scattering angle, $\phi = 45^\circ$

$\therefore$ Wavelength of the X-rays in the direction beam,

$$\lambda' - \lambda = \frac{h}{mc} (1 - \cos \phi)$$

$$\text{or, } \lambda = \lambda' - \frac{h}{mc} (1 - \cos \phi)$$

$$= 2.2 \times 10^{-12} \text{ m} - \frac{6.626 \times 10^{-34} \text{ Js}}{9.1 \times 10^{-31} \text{ kg} \times 3 \times 10^8 \text{ ms}^{-1}} (1 - \cos 45^\circ)$$

$$= 1.49 \times 10^{-12} \text{ m}$$

$$= 1.49 \text{ \AA}$$

$\therefore$ Wavelength of the X-ray is, $1.49 \text{ \AA}$

3 An X-ray photon whose initial frequency was 1.5×10^{19} Hz emerges from a collision with an electron with frequency of 1.2×10^{19} Hz. How much kinetic energy was imparted to the electron?

Soln:

Given that,

Initial frequency, $\nu_i = 1.5 \times 10^{19}$ Hz

Collision frequency, $\nu_f = 1.2 \times 10^{19}$ Hz

$\therefore$ kinetic energy imparted to the electron,

$$K.E = E_i - E_f$$

$$= h\nu_i - h\nu_f$$

$$= h(\nu_i - \nu_f)$$

$$= 6.626 \times 10^{-34} \text{ Js} \times (1.5 \times 10^{19} \text{ Hz} - 1.2 \times 10^{19} \text{ Hz})$$

$$= 1.9 \times 10^{-15} \text{ J}$$

$\therefore$ kinetic energy, $K.E = 1.9 \times 10^{-15} \text{ J}$

4. An X-ray photon of initial frequency $3.0 \times 10^{19} \text{ Hz}$ collides with an electron and is scattered through 90° . Find its new frequency.

Soln: Given that,

Initial frequency, $\nu = 3.0 \times 10^{19} \text{ Hz}$

Scattering angle, $\phi = 90^\circ$

We know,

$$\nu = \frac{c}{\lambda}$$

$$\text{or, } \lambda = \frac{c}{\nu}$$

$$= \frac{3 \times 10^8}{3 \times 10^{19}}$$

$$= 1 \times 10^{-11} \text{ m}$$

Again, We know that the Compton shift of wavelength is,

$$\lambda' - \lambda = \frac{h}{mc} (1 - \cos \phi)$$

$$\text{or, } \lambda' = \lambda + \frac{h}{mc} (1 - \cos \phi)$$

$$= 1 \times 10^{-11} + \frac{6.626 \times 10^{-34}}{9.1 \times 10^{-31} \times 3 \times 10^8} (1 - \cos 90^\circ)$$

$$= 1.24 \times 10^{-11} \text{ m}$$

$\therefore$ New wavelength, $\lambda' = 1.24 \times 10^{-11} \text{ m}$

$$\text{Again, } \nu' = \frac{c}{\lambda'}$$

$$= \frac{3 \times 10^8 \text{ ms}^{-1}}{1.24 \times 10^{-11} \text{ m}}$$

$$= 2.42 \times 10^{19} \text{ Hz}$$

$\therefore$ New frequency, $\nu' = 2.42 \times 10^{19} \text{ Hz}$

5. At what scattering angle will incident 100 keV x-rays leave a target with an energy of 90 keV?

Soln: Given that,

Initial energy of photon, $E_1 = 100 \text{ keV}$
 $= 100 \times 10^3 \text{ eV}$

$$\therefore \lambda_i = \frac{hc}{E_1}$$

$$= \frac{6.626 \times 10^{-34} \text{ Js} \times 3 \times 10^8 \text{ ms}^{-1}}{100 \times 10^3 \times 1.6 \times 10^{-19} \text{ J}}$$

$$= 1.24 \times 10^{-11} \text{ m}$$

Energy of Photon, $E_2 = 90 \text{ keV} = 90 \times 10^3 \text{ eV}$

$$\therefore \lambda_p = \frac{hc}{E_2}$$

$$= \frac{6.626 \times 10^{-34} \text{ Js} \times 3 \times 10^8 \text{ ms}^{-1}}{90 \times 10^3 \times 1.6 \times 10^{-19} \text{ J}}$$

$$= 1.38 \times 10^{-11} \text{ m}$$

$$\therefore \Delta \lambda = \lambda_p - \lambda_i$$

$$= (1.38 \times 10^{-11} - 1.24 \times 10^{-11}) \text{ m}$$

$$= 1.4 \times 10^{-12} \text{ m}$$

Again,

we know that,

$$\lambda_p - \lambda_i = \frac{h}{mc} (1 - \cos \phi)$$

$$\text{or, } \Delta \lambda = \frac{h}{mc} (1 - \cos \phi)$$

$$\text{or, } \cos \phi = 1 - \frac{\Delta \lambda mc}{h}$$

$$= 1 - \frac{1.4 \times 10^{-12} \text{ m} \times 9.1 \times 10^{-31} \text{ kg} \times 3 \times 10^8 \text{ ms}^{-1}}{6.626 \times 10^{-34} \text{ Js}}$$

$$= 0.42$$

$$\therefore \phi = \cos^{-1}(0.42)$$

=

$$\therefore \text{Scattering angle, } \phi = 65.16^\circ$$

Topic: X-ray

1. What voltage must be applied to an X-ray tube for it to emit X-rays with a minimum wavelength of 30 pm?

Soln:

Given that,

$$\begin{aligned}\text{wavelength, } \lambda_{\min} &= 30 \text{ pm} \\ &= 30 \times 10^{-12} \text{ m}\end{aligned}$$

We know,

$$\lambda_{\min} = \frac{1.24 \times 10^{-6} \text{ Vm}}{V}$$

$$\begin{aligned}\text{Or, } V &= \frac{1.24 \times 10^{-6} \text{ Vm}}{\lambda_{\min}} \\ &= \frac{1.24 \times 10^{-6} \text{ Vm}}{30 \times 10^{-12} \text{ m}} \\ &= 41333.33 \text{ V} \\ &= 4.1 \times 10^4 \text{ V}\end{aligned}$$

$\therefore 4.1 \times 10^4 \text{ V}$ must be applied to an X-ray

2: Electrons are accelerated in television tubes through potential differences of about 10 kV. Find the highest frequency of the electromagnetic waves emitted when these electrons strike the screen of the tube. What kind of waves are these?

Soln:

Given that,

$$\begin{aligned}\text{Potential differences, } V &= 10 \text{ kV} \\ &= 10 \times 10^3 \text{ V}\end{aligned}$$

We know,

$$\text{Charge of electron, } e = 1.6 \times 10^{-19} \text{ C}$$

Now,

$$\begin{aligned}v &= \frac{eV}{h} \\ &= \frac{1.6 \times 10^{-19} \times 10 \times 10^3}{6.626 \times 10^{-34}} \\ &= 2.41 \times 10^{18} \text{ Hz}\end{aligned}$$

$\therefore$ These electromagnetic waves are in the range of X-ray.